CYCLIC KUMMER EXTENSIONS: A Z/NZ GALOIS EXTENSION WITH ENOUGH ROOTS OF UNITY IS GENERATED BY AN N-TH ROOT

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The chapter prefaces this lemma with the injective character Gal(L/K) -> mu_n(K), sigma -> sigma(b)/b, and calls Kummer theory the converse; neither is part of the counted unit.

Editorial note. The label lemma-Kummer names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Same chapter: lemma-independence-characters (delivered card L48335; distinct characters are linearly independent), theorem-galois-theory (delivered card L48341; L^{Gal(L/K)} = K for a finite Galois extension), lemma-finite-Galois (|G| = [L : K] for finite Galois); linear algebra: a root of the minimal polynomial of an operator is an eigenvalue when the minimal polynomial splits with the root simple

Source reference. Original reference 0CKL: sources/stacks-fields/pinned-original-source.tex, source lines 1676–1676. Prerequisite pointer invoked at 3461; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1 (Kummer extensions)**.** *Let L/K be a Galois extension of fields whose Galois group is $\mathbf{Z}/n\mathbf{Z}$. Assume moreover that the characteristic of K is prime to n and that K contains a primitive nth root of 1. Then $L = K[z]$ with $z^n \in K$.*

Proof. Let $\zeta \in K$ be a primitive nth root of 1. Let σ be a generator of $\mathrm{Gal}(L/K)$. Consider $\sigma : L \to L$ as a K-linear operator. Note that $\sigma^n - 1 = 0$ as a linear operator. Applying linear independence of characters (Lemma 0CKL), we see that there cannot be a polynomial over K of degree $< n$ annihilating σ. Hence the minimal polynomial of σ as a linear operator is $x^n - 1$. Since ζ is a root of $x^n - 1$ by linear algebra there is a $0 \neq z \in L$ such that $\sigma(z) = \zeta z$. This z satisfies $z^n \in K$ because $\sigma(z^n) = (\zeta z)^n = z^n$. Moreover, we see that $z, \sigma(z), \ldots, \sigma^{n-1}(z) = z, \zeta z, \ldots \zeta^{n-1} z$ are pairwise distinct which guarantees that z generates L over K. Hence $L = K[z]$ as required. □

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